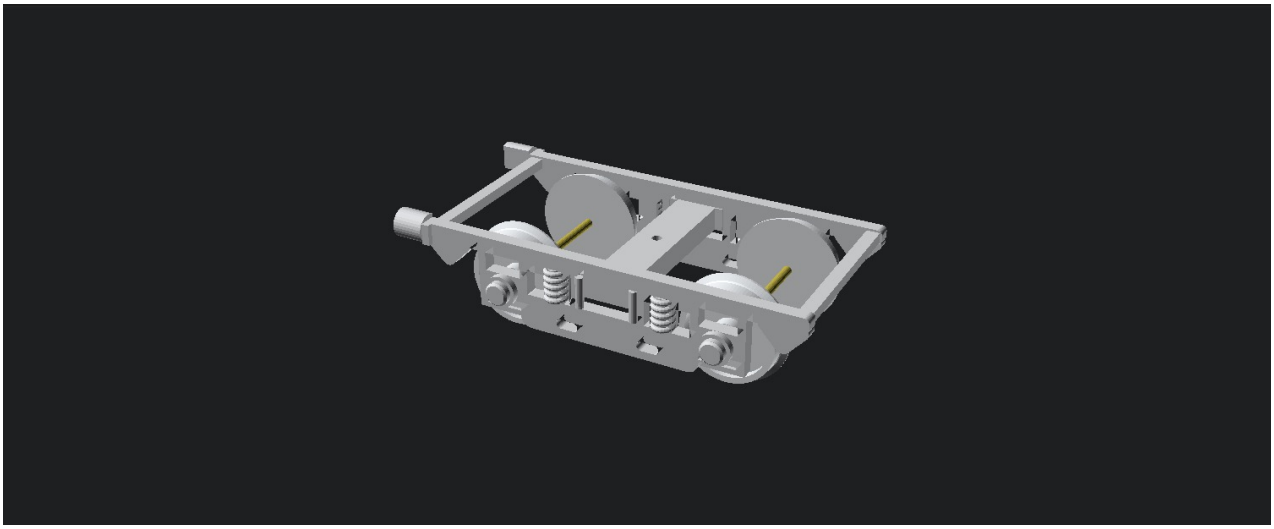
How to build - Four-wheel truck

Milwaukee Road 1947-48 passenger cars - G scale 1:29 - drawing A-4778 equalizer truck

Created by Tom Fleming, n9szf@mac.com - 4 October 2026



The finished truck: light gray frame, light silver wheels, brass axles. It has no brake rigging and no track power pickup. Each car needs two trucks.

Printed parts (per truck)

Part	Qty	File	Size, mm	On the bed
Bearing test coupon	1, once	MR_Truck4W_G_bearing_coupon.stl	36 x 12 x 5.3	pockets up
Sideframe	2	MR_Truck4W_G_sideframe.stl	140.6 x 29.8 x 9.3	inner face down; **mirror one copy**
Bolster	1	MR_Truck4W_G_bolster.stl	19.3 x 63.4 x 7.9	flat
End beam	2	MR_Truck4W_G_end_beam.stl	4.4 x 63 x 3.5	flat
Wheel	4	MR_Truck4W_G_wheel.stl	35.5 dia x 7.5	flange face down

Files are in *Fleet export / Trucks / Four-wheel truck (1947-48 cars) / G scale 1-29 / Bambu P1S / STL*. There is one sideframe file: the left and right frames are mirror images, so mirror one copy in the slicer.

Bought parts (per truck)

Item	Qty	Notes
F683ZZ flanged ball bearing	4	3 mm bore, 7 mm outside, 3 mm wide, 8.1 mm flange
3 mm solid brass round rod	2 pieces	cut each to 65.3 mm (about 131 mm of rod per truck). Metric 3 mm, not 1/8 in
M3 screw	1	truck pivot, up through the bolster into the car's bolster pad
CA glue with activator, or 2-part epoxy	-	PETG does not solvent-weld
Light silver paint	-	for the wheels

Printer settings (Bambu P1S, PETG)

Starting points, in line with the fleet's PETG settings. The truck has not been test-printed, so adjust after the first one.

Setting	All parts	Notes
Filament	PETG, dried	frame parts light gray; wheels any colour (painted)
Nozzle / plate	0.4 mm nozzle, textured PEI plate	no glue stick needed on textured PEI
Temperatures	240-250 C nozzle, 70-80 C bed	same as the car bodies
Supports	none	every part has a flat face for the bed
Brim	off; 3-5 mm on the end beams if they lift	end beams are thin and narrow

Part	Layer	Walls	Infill	Notes
Bearing test coupon	0.12 mm	4	100%	print it with the same settings as the sideframes, or the fit will not match
Sideframe	0.12 mm	4	40%	inner face down; the bearing pockets open on the bed face - slow first layers, check for elephant foot
Bolster	0.2 mm	4	40%	flat
End beam	0.2 mm	4	100%	flat; small enough to be solid
Wheel	0.12 mm	4	100%	flange face down; slow outer wall for a round tread; seam set to Aligned, then sand the seam line off the tread

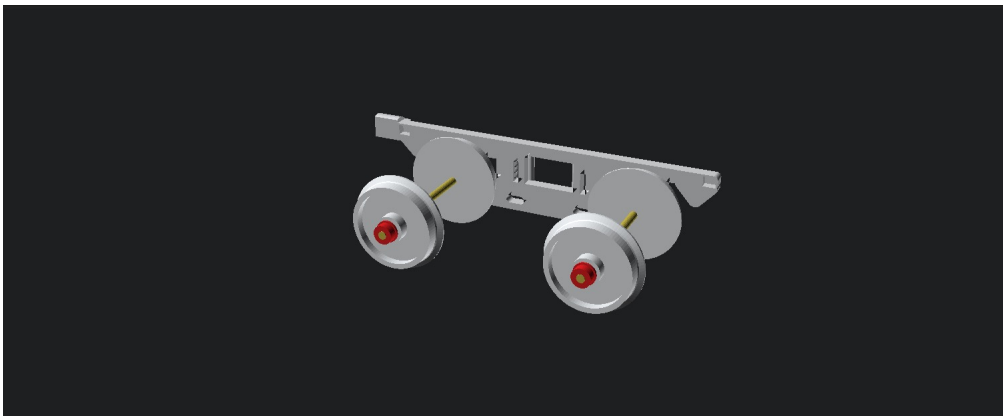
Printing

1. Print the frame parts (sideframes, bolster, end beams) in **light gray PETG** on the Bambu P1S. No supports are planned for any part.
2. Print the wheels in PETG, flange face on the bed. Any colour will do, as they get painted.
3. **Print the bearing test coupon first.** It has three pockets marked **-**, **0** and **+** (0.05 mm apart). Press a bearing into each. If the **0** pocket holds the bearing firmly without cracking, carry on. If **-** or **+** is better, change BEARING_FIT in common/mr_config.scad by 0.05 and remake the sideframe STL before printing the sideframes.

Getting the parts ready

1. Clean any stringing out of the bearing pockets, the bolster slot and the two end beam slots on the inner face of each sideframe.
2. Cut **two** brass axles to **65.3 mm** for each truck. File the ends square and put a small chamfer on them.
3. Paint the four wheels light silver and let them dry. Keep paint out of the bore and off the tread if you can.
4. Test one wheel on a scrap of the brass rod. The bore is printed slightly under 3 mm for a press fit. If it is too tight, open it with a 3 mm drill by hand; if it is loose, a drop of CA on assembly will hold it.

Assembly



The bearings (shown red) sit in the journal boxes; each brass axle runs right through its two bearings.

1. **Bearings.** Press one F683ZZ bearing into each journal box pocket from the inner face of the sideframe, flange last, until the flange is seated in its recess. Two per sideframe.
2. **Wheelsets.** Press two wheels onto each axle with the flanges facing each other (toward the middle). Set them **40.5 mm back-to-back** (between the inside faces of the flanges) and centred, so the same length of axle sticks out each side.
3. **First sideframe.** Lay one sideframe inner face up. Glue one end of the bolster into the centre slot, and one end of each end beam into the slot at each end of the frame. Check they stand square and let the glue set.
4. **Fit the wheelsets.** Slide one end of each axle into its bearing on that sideframe.
5. **Second sideframe.** Put glue in the bolster slot and the two end beam slots of the other sideframe, then lower it on, guiding the two axle ends into its bearings. Keep the brake cylinder ends of both sideframes at the same end of the truck.
6. **Square it up.** Stand the truck on a flat surface (or a piece of track) while the glue sets, so all four wheels touch.
7. **Check.** Each wheelset should spin freely with a little side play (about 0.4 mm). If an axle binds, the sideframes are pinched in or the axle is too long.
8. **Fit to the car.** Fix the truck under the car's bolster pad with the M3 screw through the hole in the bolster. Leave it just loose enough to swing freely. The brake cylinder end can face either way.

Notes

This truck design has been checked in OpenSCAD only and has not been test-printed yet, so treat the fits (bearing pocket, wheel bore, glue slots) as first guesses and adjust after the first build.